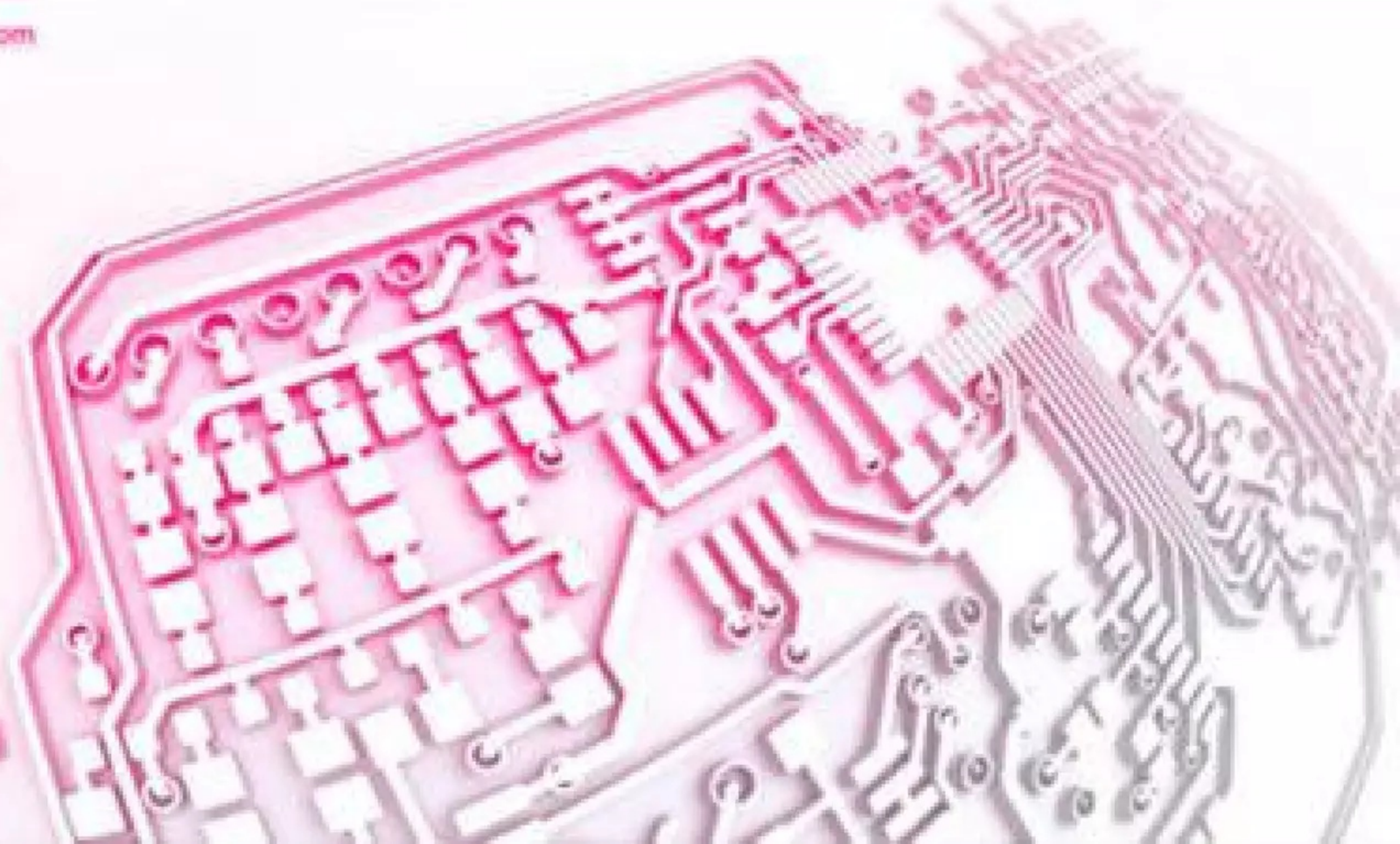
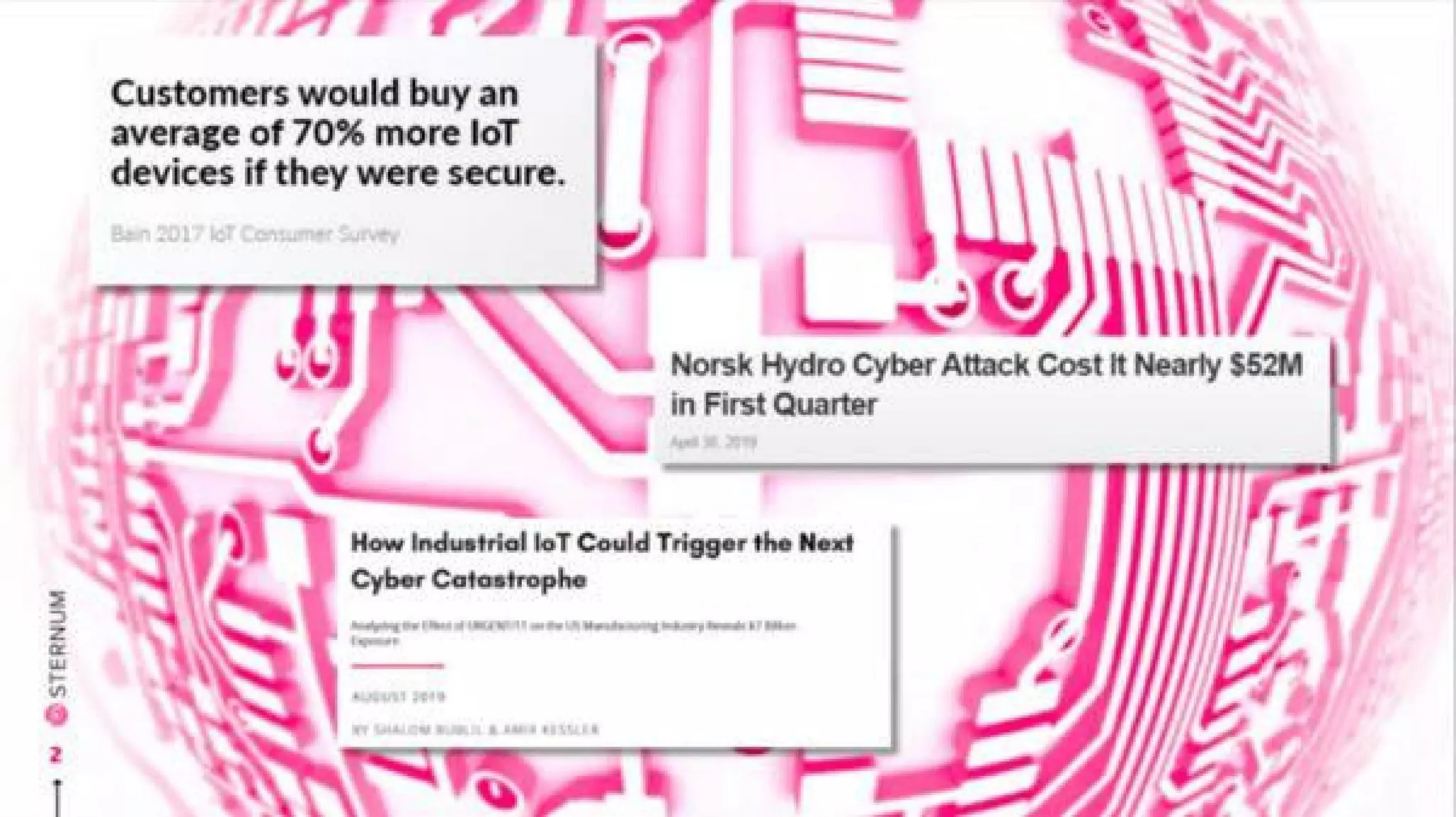
STERNUM

Securing IoT Devices, From Within

Natali Tshuva | CEO | natali@sternumiot.com
Boaz Shedletsky | CBO | boaz@sternumiot.com




Customers would buy an average of 70% more IoT devices if they were secure.
Bain 2017 IoT Consumer Survey
Norsk Hydro Cyber Attack Cost It Nearly $52M in First Quarter
How Industrial IoT Could Trigger the Next Cyber Catastrophe
AUGUST 2019

June 16
WIRED

A Legion of Bugs Puts Hundreds of Millions of IoT Devices at Risk

The so-called Ripple20 vulnerabilities affect equipment found in data centers, power grids, and more.

STATUS: CONFIRMED (8)
B. Braun
Baxter
Caterpillar
HP
Intel
Maxlinear (through HLFN)
Rockwell
Sandia National Labs
Schneider Electric/APC
Digi
HCL Tech

STATUS: NOT AFFECTED (5) (as reported by vendor)
Abbott (through Guidant Healthcare)
Amd
GE Healthcare
Laird
Zebra Technologies

STATUS: PENDING (19)		
Cisco (through Starent)	Hitachi europe	Synamedia(Through Cisco)/NDSUK
EMC (now Dell)	Hilti	Syncroness
GE general electric (through quadros)	Honeywell	Technicolor (Throught CISCO)
NASA	Itron	Texas Instruments-Berlin
Philips	Kadak	Thinkcom/ThinKam
Verifone	L-3 Chesapeake Sciences Corporation	Tollgrade communications
	Lockheed martin	Ultra Electronics Flightline Systems
**	Maxim Integrated Products	Vicom
Agilent	Memjet	Videotek
Airbiq(throug Netsnapper Technologies SARL)	MTS Technologies	Vocera
Arburg	Netafim	vpacket(now DASAN Zhone)
BAE systems	Netsnapper Technologies SARL	Weibel weibel.dk
BD	NVIDIA (through portalplayer)	Western geco
Broadcom	Portalplayer	Xilinx
Capsule (through digi)	Qualstar.com	Zodiac Aerospace
DASAN Zhone(through vpacket)	Red lion controls	Quadros
Datamax Corporation	Redcom	BECK
Enghouse (through tollgrade communications)	SAIC	KADAK (defunct)
Foundry	ScriptPro	Texas Instruments

ALL BUSINESSES FACE IoT SECURITY RISKS

>300%

Increase of Attacks on IoT in 2020

>65%

Of CISOs Unconfident to assess & mitigate IoT risk

>50%

CISOs plan to increase staffing & spending

84%

of businesses insufficiently prepared for IoT security

Source: McKinsey & Co.

STERNUM

Source: Gartner, 2018 IoT Security Survey Report



Sternum secures **IoT devices** by preventing cyber attacks **on-device**, with **real-time monitoring** and **protection**, the **lowest overhead**, and **unique visibility** into devices

Sternum puts an end to the endless race of vulnerability patching by ensuring embedded protection for every IoT device – **new, old, existing, low or high end, Linux or RTOS – proven on 5-year-old CENSORED**

Notable Partners

CENSORED

Télit

Winner of Next Gen Cybersecurity IoT Solution During RSA 2020



THE TEAM

Offensive and defensive **cyber experts,** who are deeply experienced in low-level embedded systems.



Natali Tshuva
CEO



Boaz Shledetsky
CBO



Liam Granot
VP R&D



Arik Farber
Head of Research

Unit 8200

Kellogg

Cellebrite

WHY IS IoT SO DIFFICULT TO SECURE?

High Diversity & No standardization
Operating Systems I Communication Protocols I Hardware

Low Confidence & 3rd party dependencies
3rd Party Libraries I Communication Modules | Homegrown Code

Resource-constrained
Compute | Memory | Battery | Bandwidth

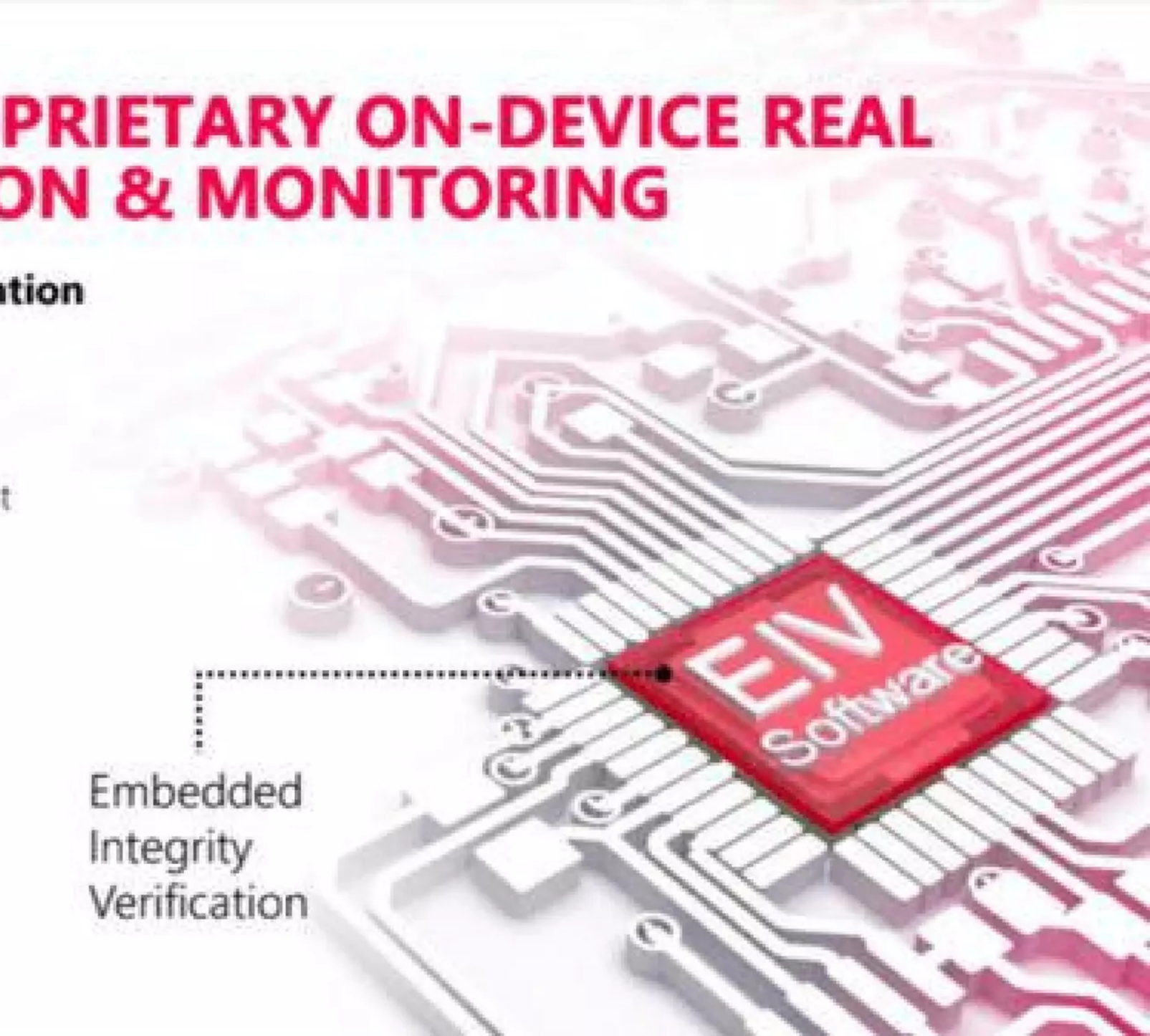
IoT worldwide is growing exponentially,
but security solutions are 20 years behind



SOLUTION: PROPRIETARY ON-DEVICE REAL TIME PROTECTION & MONITORING

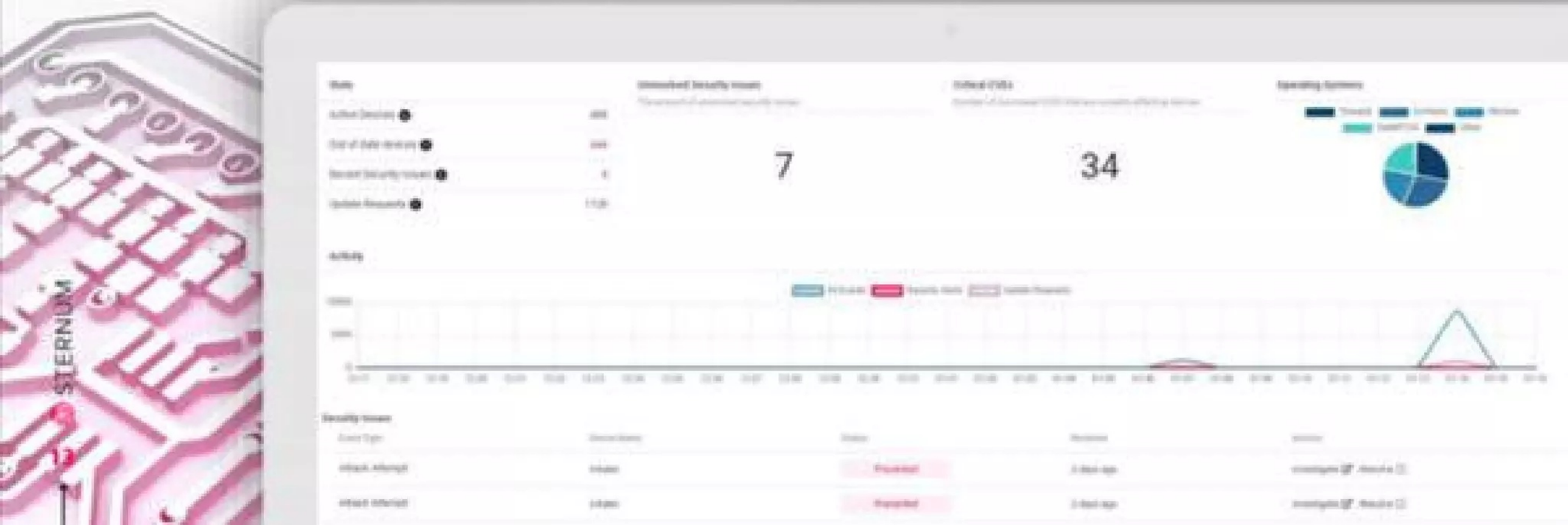
Integrity-based attack prevention

- Using patent-pending EIV (3 Patents Pending)
- Protects exploitable points without patching vulnerabilities
- Minimal overhead
- Works for both source code and 3rd party code automatically
- Seamlessly integrated with both Linux & RTOS IoT devices



ADS: DETECTION, MONITORING & ASSET MGMT

- Access to **unique data**: visibility and management within IoT/Edge devices + 3rd party code
- **Immediate alerts** on attack attempts and detected anomalies
- Advanced IoT **analytics**: Investigates cyber-breaches easily with intuitive story-based timeline



STERNUM

Securing IoT Devices, From Within

Thank you!

Natali Tshuva | CEO | natali@sternumiot.com

Boaz Shedletsky| CBO | boaz@sternumiot.com

